DAY-1

Wireshark and OSI Model Assignment

Q1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Answer:

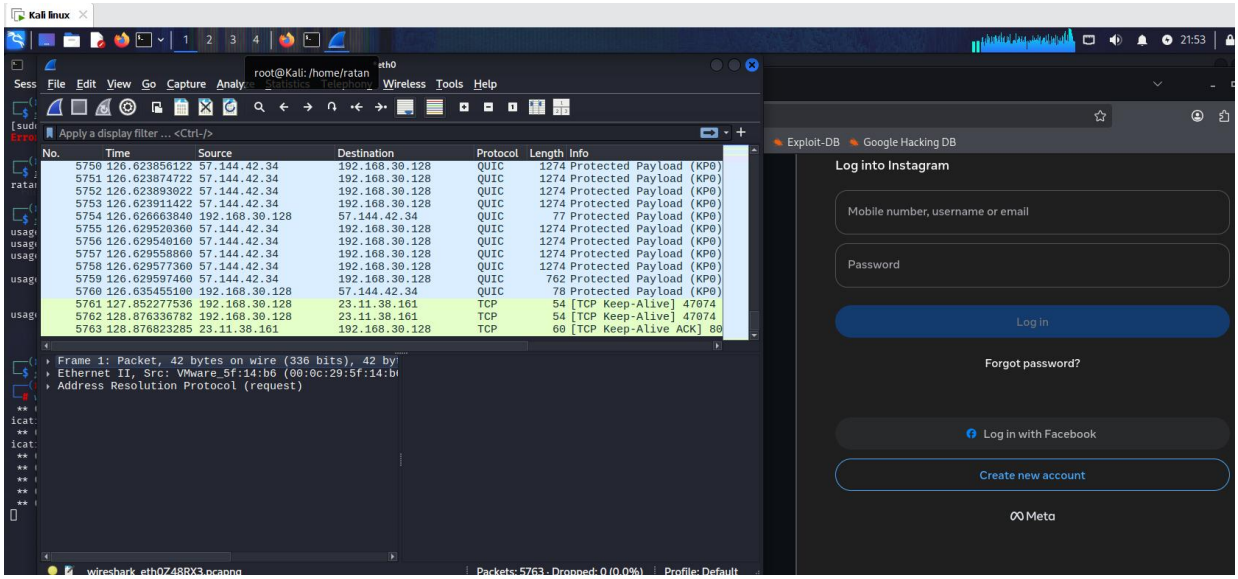
1. **Install:** Wireshark 4.6.6 was already installed in my Kali Linux.
2. **Start capture:** Open Wireshark and double-click the main interface eth0 (the one with a moving graph line). The capture starts.
3. **Visit a website:** Open Firefox in Kali and go to instagram.com.
4. **Stop capture:** Click the red Stop button in Wireshark.
5. **Note protocols:** Look at the Protocol column of the packet list and write down the names. Filters like dns, tcp or tls help to find them.

Protocols normally seen when opening a website:

Protocol	What it does	OSI layer
DNS	Finds the IP address of the website name (e.g. instagram.com)	Application (7)
TLSv1.2 / TLSv1.3	Encrypts the data between browser and website (HTTPS)	Presentation (6)
TCP	Makes a reliable connection and delivers data in order	Transport (4)
UDP / QUIC	Fast transport used by modern websites and video	Transport (4)
HTTP	Web page request (seen mainly on non-HTTPS sites)	Application (7)
ARP	Finds the MAC address of the router on the local network	Data Link (2)
ICMPv6 / mDNS / SSDP	Background messages from devices on the network	Network (3) / Application (7)

Note: Most sites now use HTTPS, so you will mostly see TLS (or QUIC) and only a little plain HTTP.

Protocols I saw in my capture: ARP, DNS, TCP, TLSv1.3, QUIC and HTTP (the capture had 5763 packets).



Wireshark packet list of the capture on eth0 (instagram.com opened in Firefox)

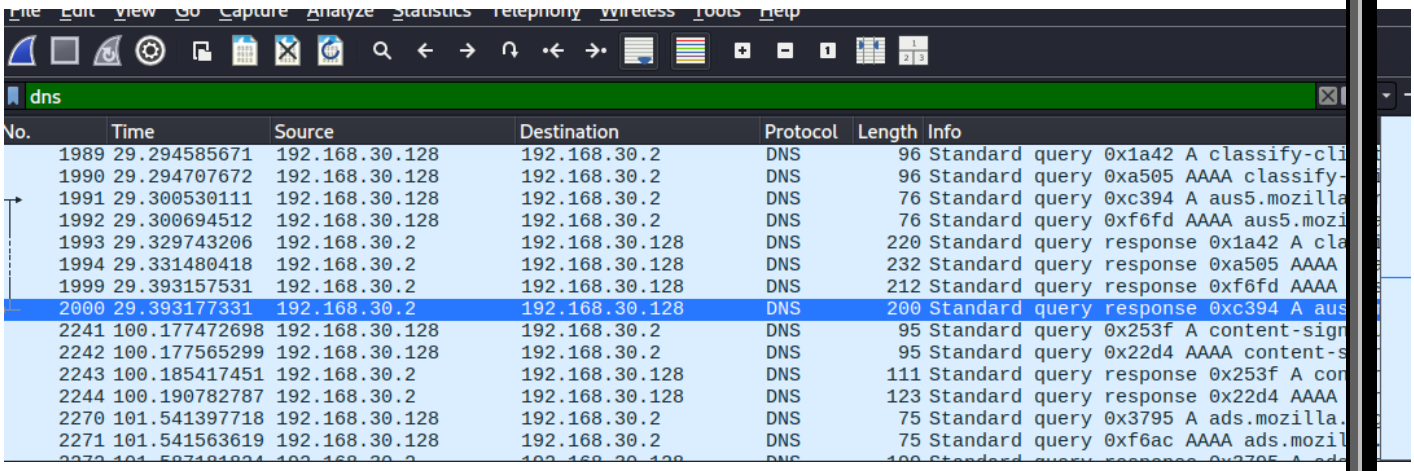
Q2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Answer:

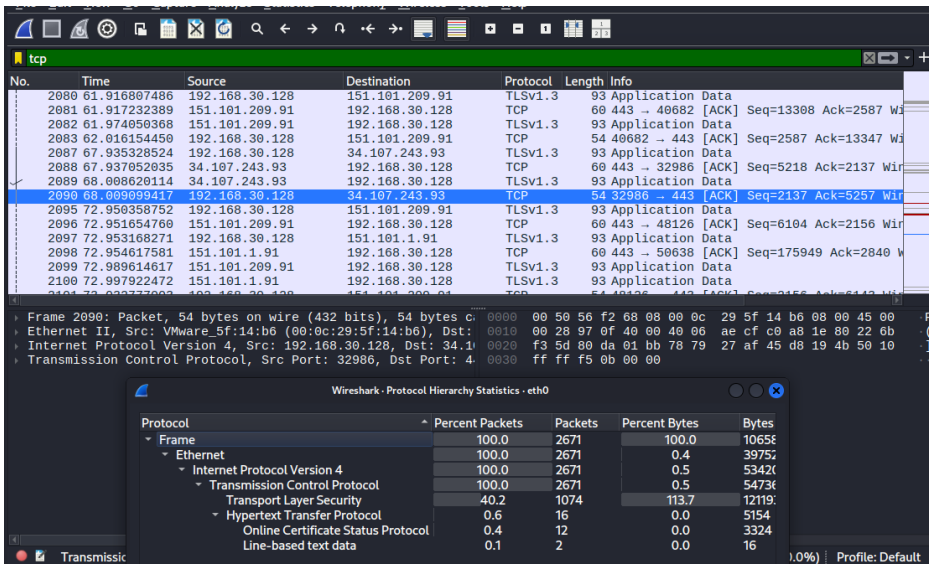
Three packets from my capture and the OSI layer each one mainly represents:

Packet no.	Protocol	Info column (typical)	OSI layer	Reason
2000	DNS	Standard query response 0xc394 A aus5.mozilla.org	Application (7)	DNS is a service used by the browser to turn a website name into an IP address.
2090	TCP	32986 > 443 [ACK] Seq=2137 Ack=5257	Transport (4)	It shows port numbers (443 = HTTPS) and the ACK that confirms data was received.
1	ARP	Who has 192.168.30.2? Tell 192.168.30.128	Data Link (2)	ARP works with MAC addresses inside the local network (Ethernet frame).

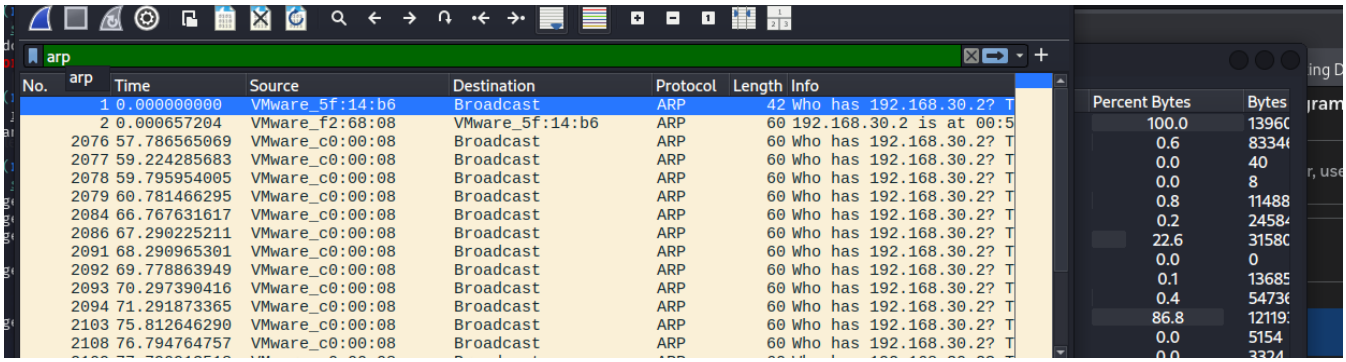
Quick mapping: HTTP, DNS = Application (7); TLS = Presentation (6); TCP, UDP = Transport (4); IP, ICMP = Network (3); ARP, Ethernet/Wi-Fi frames = Data Link (2).



Packet 2000: DNS (filter: dns)



Packet 2090: TCP (filter: tcp)



Packet 1: ARP (filter: arp)

Q3. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

Answer:

OSI MODEL WITH DAILY APP EXAMPLES

Layer	Name	Example from apps I use daily
7	APPLICATION	WhatsApp, Zomato and Chrome send/receive using HTTPS
6	PRESENTATION	Instagram photos (JPEG) and WhatsApp end-to-end encryption
5	SESSION	Staying logged in on Instagram, a Google Meet call session
4	TRANSPORT	TCP: WhatsApp messages (reliable) UDP: PUBG / video calls (fast)
3	NETWORK	IP address + routers find the path to the Instagram server
2	DATA LINK	Wi-Fi frames using MAC address between phone and router
1	PHYSICAL	Wi-Fi radio waves, mobile tower signal, fiber cable

Sending: Layer 7 to Layer 1 **Receiving: Layer 1 to Layer 7**

Q4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Answer:

When I type "Hello" and press send in WhatsApp, the message goes down through the layers on my phone, travels across the network, and goes up through the layers on my friend's phone.

- **Application layer (7):** I type the message in the WhatsApp app and press send. The app creates the message data and prepares to send it to WhatsApp's server.
- **Presentation layer (6):** The message is encrypted (WhatsApp end-to-end encryption) so that nobody on the way can read it. Photos or videos would also be compressed here.
- **Session layer (5):** The connection (session) between my phone and the WhatsApp server stays open, so the message can be sent and the delivery ticks come back.
- **Transport layer (4):** The message is split into small segments and numbered. TCP adds port numbers and makes sure all pieces arrive correctly, and re-sends any lost piece.
- **Network layer (3):** IP adds my phone's IP address and the WhatsApp server's IP address. Routers on the Internet use these addresses to choose the path.
- **Data Link layer (2):** The data is put into frames with MAC addresses so that it can move from my phone to the Wi-Fi router.
- **Physical layer (1):** The bits are sent as radio waves (Wi-Fi or mobile data), and then as electrical or light signals in cables.

Receiving: The WhatsApp server passes the message on to my friend's phone. There the layers work in reverse order, from layer 1 up to layer 7: the signals become bits, frames, packets and segments, the message is joined and decrypted, and it appears as "Hello" on the screen.